

REDAD: A Reliable Distillation for Image Anomaly Detection

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Abstract—The vanilla knowledge distillation (KD)-based approach for image anomaly detection (AD) encounters three main challenges: data shift, normality forgetting, and anomaly overgeneralization. To effectively address these challenges, we introduce REDAD, a reliable distillation strategy for AD, which avoids intricate network designs. Initially, we incorporate an adaptive teacher model (ATM) that dynamically adjusts the teacher's batch normalization (BN) statistics during the distillation process, aligning the teacher with the target training data distribution. Next, we introduce a normality remembering enhancement (NRE) module, designed to compel the student to learn the most challenging normal feature with high distillation loss, thereby bolstering normality retention and reducing false positives. Finally, we present a novel direction-guided regularization (DGR) technique to robustly enlarge the divergence in abnormal feature pairs, preventing the oversight of abnormal regions, or false negatives. Comprehensive experiments on the MVTec, VisA, and MVTec3D datasets show that REDAD effectively resolves these three concerns, achieving superior AD performance compared with its baseline model (exceeding 3.5% (5.0%, 6.1%) in I-AUROC, 2.3% (1.6%, 1.5%) in P-AUROC, and 3.2% (5.4%, 3.8%) in PRO). In addition, two real-world industrial product inspection applications further underscore the efficacy and utility of the proposed REDAD method.

Index Terms—Knowledge distillation (KD), self-supervised learning, unsupervised anomaly detection (UAD).

I. INTRODUCTION

THE objective of image anomaly detection (AD) is to identify images that are irregular and precisely locate the specific areas that deviate from normal patterns. This technique for detecting anomalies is used across multiple sectors, including industrial inspection [1], [2], medical diagnostics [3], [4], and video surveillance [5]. However, in real-world applications, the AD tasks often encounter challenges posed by the scarcity of abnormal samples and the vast diversity of anomaly types. To address this issue, unsupervised AD (UAD) methods have emerged as a preferred solution as they minimize the need for extensive annotated data [6].

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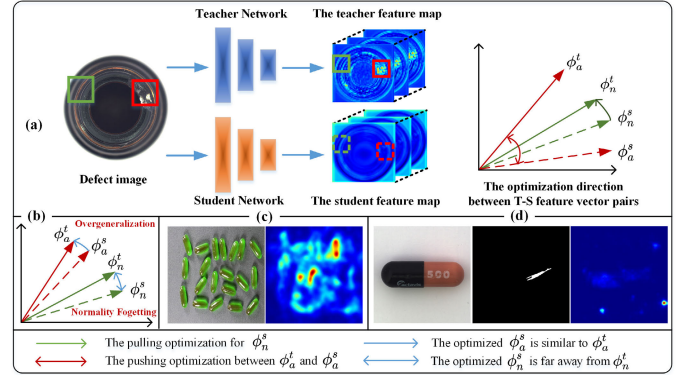


Fig. 1. Issues in KDST-based AD. (a) General KDST method framework. (b) Illustration highlighting normality forgetting and overgeneralization problems. (c) Example of normality forgetting with a normal sample. (d) Example of anomaly overgeneralization with an abnormal sample.

Among various UAD methods, knowledge distillation (KD) [7]-based methods establish a student-teacher (S-T) network pairing to detect anomalies by pinpointing the feature-to-feature discrepancies [8], as shown in Fig. 1(a). During the training phase, the teacher network, pretrained on an image classification dataset like ImageNet, remains unchanged, while the student network is trained exclusively using normal images from the AD task. In the testing phase, the student network is expected to mimic the teacher in normal feature regions but diverge in abnormal regions, and this divergence is then used to identify anomalies. Typically, cosine similarity is used as a metric to gauge feature discrepancies, owing to its precision in capturing relationships in both high- and low-dimensional data [9]. The remarkable knowledge transfer and learning guidance offered by KDST-based methods have propelled them to the forefront of performance in the field [6]. Beyond the standard configuration, methods such as STPM [10] and MKD [11] enhance their effectiveness by incorporating multi-scale features from various network layers into the distillation process, thereby improving overall performance.

Despite the promising results of KDST-based methods, they encounter three main issues: data shift, normality forgetting, and anomaly overgeneralization. Regarding data shift, the distribution of images in the AD task typically differs from that of ImageNet. As the data distribution evolves in the new task, the BN statistics in the student and teacher models rapidly diverge, resulting in increased KD loss [12]. This divergence hinders the alignment of features between the teacher and student models. With respect to the issue of normality

forgetting, the student model, being trained on amounts of anomaly-free data, becomes sensitive to the nuanced patterns in normal data due to the long-tail problem. This sensitivity results in discrepancies in the normal feature representations between the teacher and student models [refer to Fig. 1(b)], thereby increasing the likelihood of false positives in AD [refer to Fig. 1(c)]. To address this, MemKD [13] proposes a normality recall memory to reinforce the normality in student-generated features, ensuring detailed patterns are accurately captured. EfficientAD [14] introduces a hard feature loss that focuses backpropagation on the output elements with the highest loss, reducing false-positive detections on normal images.

Anomaly overgeneralization arises when the student model, due to its strong generalization capabilities, produces abnormal feature distributions similar to those in the teacher's domain [refer to Fig. 1(b)] [15], [16]. This overgeneralization leads to the oversight of abnormal regions [refer to Fig. 1(d)]. To tackle this problem, various strategies have been devised. Both AST [17] and ATSN [18] use an asymmetric student-teacher structure to widen the gap in anomaly representations between teacher and student. On the other hand, reliable direction (RD) [19] introduces a reverse distillation approach featuring a teacher encoder and a student decoder, fostering diversity in anomalous representations. A network structure similar to RD is also proposed in [4]. Some other works suggest that using only normal data for training can lead to low sensitivity to abnormal samples. Therefore, they use self-supervised methods by simulating anomalies to mitigate overgeneralization. With the help of synthetic anomalies, CDO [15] collaboratively minimizes discrepancies in normal feature distributions and maximizes those in synthetic abnormal feature distributions, which achieves remarkable performance. Operations similar to CDO are also proposed in Pull&Push [16]. From synthetically corrupted normal images, DeSTSeg [20] trains a denoising student network to match the teacher network feature of the same images without corruption. Therefore, the two networks are reinforced to generate distinct features from anomalous inputs.

This study proposes a reliable distillation strategy within the pyramid KDST framework to simultaneously addressing the three aforementioned issues. Inspired by exemplar-free class incremental learning method [12], we introduce an adaptive teacher model (ATM) to tackle the data shift problem. This ATM updates the BN statistics of the teacher model while keeping the rest of the model parameters fixed during the training process, significantly minimizing KD loss. To counter the normality forgetting problem, we propose a normality remembering enhancement (NRE) module, influenced by the online hard example mining (OHEM) technique [21]. Unlike MemKD [13], which uses a memory-intensive mechanism to retain normal information, NRE effectively directs the student model to concentrate on more complex normal features by identifying those with highest loss. Furthermore, unlike EfficientAD [14], which uses a static mining factor, NRE adopts a gradually decreasing factor to ensure training stability.

To tackle the challenge of anomaly overgeneralization, we devise a direction-guided regularization (DGR) technique

with synthetic anomalies. Formally, let the normal feature vector pairs for the student and teacher be denoted as (ϕ_n^s, ϕ_n^t) , which are trained to align closely, the objective of DGR is to augment the disparity between the perturbed abnormal feature pair (ϕ_a^s, ϕ_a^t) . The DGR mechanism achieves this by adjusting ϕ_a^s away from ϕ_a^t in a mirrored direction when the cosine angle between ϕ_a^t and ϕ_n^t is positive, and in the opposite direction when negative. We call this method as RD, as it effectively addresses the weaknesses of weak discrimination or instability found in existing methods such as DeSTSeg [20] and CDO [15], enabling the detection of both subtle and pronounced anomalies. A more comprehensive description and analysis can be found in Section III-G. It is crucial to note that DGR does not require additional network architectures like those mentioned in the aforementioned methods [17], [19], [20], thus introducing no extra computational costs during testing.

The contributions of this study can be summarized as follows.

- 1) We identify and address three prevalent and independent issues within the KDST-based methods. To this end, we introduce a reliable distillation approach, termed REDAD, which can simultaneously mitigate these three challenges within the KDST framework. Furthermore, REDAD enhances the performance of the KDST-based methods without incurring additional network complexity or inference costs.
- 2) We design a novel DGR technique that effectively amplifies the divergence of anomaly features between the teacher and student models using self-supervised training with synthetic anomalies.
- 3) Extensive experiments across diverse datasets demonstrate that REDAD outperforms the current state-of-the-art KDST-based methods in both image-level and pixel-level AD tasks.

The remainder of this article is organized as follows. Section II comprehensively reviews the related works. REDAD is illustrated in detail in Section III. Section IV presents thorough experiments, results, and analyses. Finally, Section V concludes this article and discusses future research directions.

II. RELATED WORKS

A. Unsupervised AD

Most UAD techniques can be broadly categorized into two approaches: reconstruction-based and feature-embedding methods [6]. Reconstruction methods, such as autoencoders [22], GANs [23], and denoising diffusion [24], assume that anomalies in data cannot be accurately reproduced by a neural network trained only on normal data. Therefore, discrepancies in pixel-level reconstruction are used as indicators for detecting anomalies. Nevertheless, this assumption is not always correct. Occasionally, networks may generalize too much, resulting in false negatives when they mistakenly reconstruct anomalies [25].

Currently, feature embedding-based methods are demonstrating superior performance. These approaches often leverage pretrained models to extract comprehensive feature

representations. Subsequently, statistical algorithms such as multivariate Gaussian distribution [26], memory bank [27], and normalizing flow [28], [29] are used to embed the distributions of normal features. Anomalies are then identified by comparing the input features against the learned distribution or the stored clustering centers. Despite recent advances in anomaly localization, statistical algorithms often struggle with high computational complexity and memory consumption [25]. In addition, normalizing flows can suffer from training instability, compromising the flexibility of density estimation [17]. Alternatively, KD-based methods [8], [10], [15], [17], [19] have emerged as a promising approach. These methods establish a teacher–student network pair and detect anomalies by identifying discrepancies in feature representations between expert and apprentice domains, achieving state-of-the-art results in AD.

B. Self-Supervised Mask Training

Despite the absence of genuine anomalous data for training within the UAD framework, it is feasible to simulate pseudo-anomalous data. This simulation enables the training of an AD model in a self-supervised manner, thereby enabling it to explicitly grasp the potential disparities between normal and abnormal samples. Methods such as CutPaste [30] generate pseudoanomalies by adding small patches to a normal image and training a model to detect these anomalous regions. Similarly, DRAEM [31] overlays additional texture images as noise onto normal images to generate abnormal regions, focusing on creating textural anomalies. Building on DRAEM [31], DeSTSeg [20] trains a denoising student encoder–decoder using synthetically corrupted normal images, striving to align with the features of uncorrupted images captured by the teacher network. CDO [15] perturbs the input normal image with Gaussian noise and maps the perturbed image into distinct expert and apprentice domains using respective mapping functions. Alternatively, SimpleNet [25] generates synthetic anomalies within the feature space, operating under the assumption that defects may not exhibit significant commonality in the image space. In our system, we incorporate the principles of DRAEM [31] for anomaly simulation and detect synthetic anomalies within a KD framework.

C. BN Updating in KD

BN, a widely used technique in deep learning, may encounter challenges in situations where the data distribution undergoes changes. To facilitate domain transfer, various domain adaptation methods leverage normalization statistics, as cited in [32], [33], and [34]. In the context of KD, the BN statistics in both the student and teacher models can diverge rapidly, resulting in an elevation of KD loss, as discussed in [12]. Recent studies have investigated the potential of modifying the teacher model through meta-learning to enhance knowledge transfer within the conventional KD framework, as exemplified in [35]. Furthermore, research on KD, specifically [36], suggests that updating the teacher model’s normalization statistics with data used for student training can yield performance improvements. Notably, while

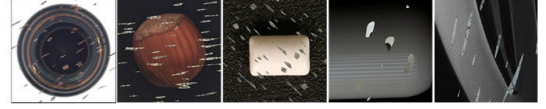


Fig. 2. Some samples of perturbed normal images.

teacher model modification has been explored in traditional KD scenarios, our approach represents the first application to AD.

III. METHOD

A. Problem Definition

For any image $x_i \in \mathbb{R}^{H \times W \times 3}$ in $\mathcal{X}_{\text{train}} \cup \mathcal{X}_{\text{test}}$, where $\mathcal{X}_{\text{train}}$ is a set of anomaly-free training images and $\mathcal{X}_{\text{test}}$ is a test query set containing both normal and abnormal samples. The goal of AD is to train a model to recognize and localize anomalies in the query set.

B. Formulation of General KDST-Based Methods

We denote the pretrained teacher network as ϕ^t and the distilled student network as ϕ^s . For an image x_i , ϕ^t and ϕ^s extract features from different hierarchies. We define $\mathcal{L}$ as the subset including the indexes of hierarchies for use. The feature map from level $l \in \mathcal{L}$ is denoted as $\phi^l \sim \phi^l(x) \in \mathbb{R}^{H_l \times W_l \times C_l}$, where H_l , W_l , and C_l are the height, width, and channel size, respectively. For an entry $\phi_{h,w}^l \in \mathbb{R}^{C_l}$ at location (h, w) , the distance between $\phi_{h,w}^{t,l}$ and $\phi_{h,w}^{s,l}$ can be formulated as $D(\phi_{h,w}^{t,l}, \phi_{h,w}^{s,l})$ where $D(\cdot, \cdot)$ denotes a function for feature-to-feature discrepancy calculation. The KD process for the AD task is to optimize $D(\phi^t, \phi^s)$ using the normal samples in $\mathcal{X}_{\text{train}}$.

Let $p \sim (h, w)$, and define p_n and p_a as the normal feature and abnormal feature positions, respectively. The general KD optimization loss can be succinctly expressed as

$$L_{\text{gkd}} = \frac{1}{N} \sum_{l=1}^{\mathcal{L}} \sum_{p_n}^{\mathcal{P}_n} (1 - \cos(\phi_{p_n}^{t,l}, \phi_{p_n}^{s,l})). \quad (1)$$

Here, the distance metric is based on cosine similarity, given by $D(\cdot, \cdot) := 1 - \cos(\cdot, \cdot)$. The KDST-based methods assume that the discrepancies between $\phi_{p_a}^t$ and $\phi_{p_a}^s$ would not be significantly affected and keep large during the KD process which merely minimizes the discrepancies between $\phi_{p_n}^t$ and $\phi_{p_n}^s$. However, because of the overgeneralization issue, the discrepancies between $\phi_{p_a}^t$ and $\phi_{p_a}^s$ may also decrease, resulting in high prediction uncertainty.

In KDST-based methods, convolutional neural networks such as ResNet and its derivatives, which incorporate BN layers, are commonly used as models for both ϕ^t and ϕ^s . During the KD process, the parameters of ϕ^t , including the BN statistics derived from pretrained datasets like ImageNet, remain unchanged, while the parameters of ϕ^s are subject to updates in the new AD task dataset. The divergence in BN statistics between ϕ^t and ϕ^s leads to an increase in KD loss, ultimately impeding the precise alignment of features between the teacher and student models.

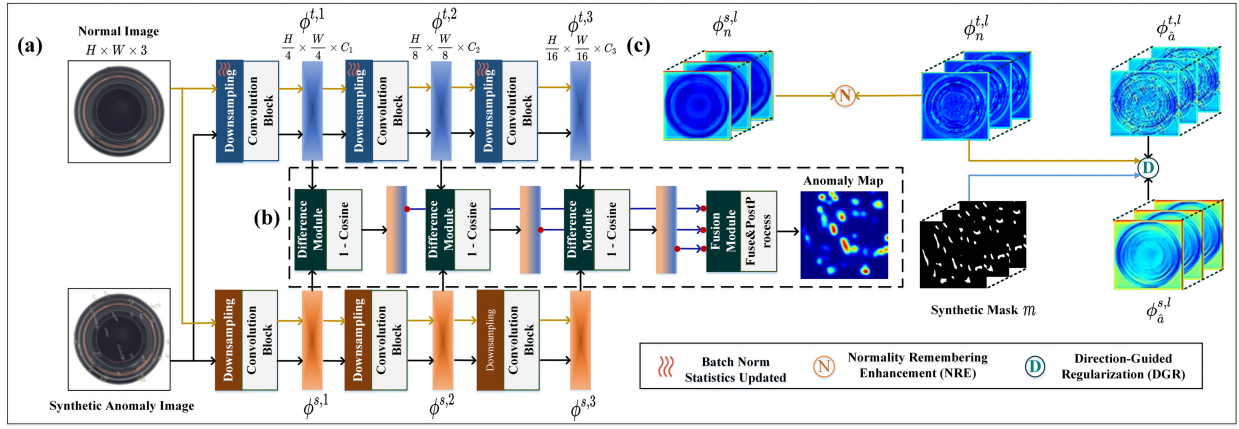


Fig. 3. Overview of REDAD. (a) REDAD uses the general KDST framework, integrating a pretrained adaptive teacher network. During training, both the teacher and student networks receive normal and synthetic anomaly images as input. (b) Module for computing anomaly maps in the inference stage. (c) Feature flows throughout the distillation training process.

C. Simulated Anomaly Generation

As mentioned before, the DGR uses synthetic anomalous images for self-supervised mask training. In this study, we adopt the same algorithm proposed in [31]. Random 2-D Perlin noise is generated and binarized by a preset threshold to obtain an anomaly mask m . An anomalous image x_a is produced by replacing the mask area with a linear combination of an anomaly-free image x_n and an arbitrary image from external data source z , controlled by an opacity factor γ , chosen at random within the interval $[0.15, 1]$. The process is mathematically represented as

$$x_a = \gamma(m \odot z) + (1 - \gamma)(m \odot x_n) + (1 - m) \odot x_n \quad (2)$$

where $\odot$ signifies elementwise multiplication. This anomaly generation occurs online during the training phase. Fig. 2 displays samples of perturbed normal images across various datasets. We can see that the anomaly mask generated by random Perlin noise is irregular, resembling actual anomalies.

D. Formulation of REDAD

The overall architecture of REDAD is depicted in Fig. 3, using the same design as general KDST-based methods without additional structural components. Distinctions arise during the distillation optimization process, highlighted by three key aspects, which address the three separate issues. First, we introduce an ATM which updates the BN statistics of the teacher model while fixed the rest model parameters during the training process. ATM allows for significantly reduced KD loss, thereby enhancing alignment of features between the teacher and student models. Second, we incorporate an NRE module, which supersedes the general feature distillation in (1) by pinpointing normal feature pairs with low cosine similarity through a dynamically decreasing mining factor. Finally, we devise a DGR technique to reliably widen the divergence in the synthetic abnormal feature regions between the teacher and student, efficiently mitigating the overgeneralization problem.

E. Adaptive Teacher Model

The ATM initially converts the pretrained teacher model into a two-class classification model using linear probing, which

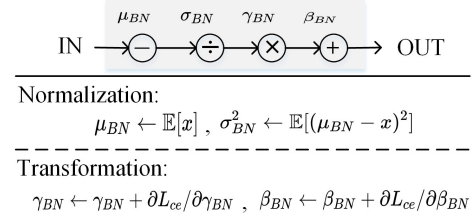


Fig. 4. ATM modulates the teacher features by estimating normalization statistics μ_{BN} and σ_{BN} and optimizing transformation parameters γ_{BN} and β_{BN} .

involves updating only the final linear layer [37]. We assign the classification label for the synthetic anomaly image x_a as 1 and that for the normal image x_n as 0. Let $y_t(x)$ represent the predicted probability of the teacher model for an input image x . The cross-entropy contrast loss is then defined as

$$L_{ce} = -\frac{1}{2N_{BN}} \sum_{i=1}^{N_{BN}} \log(y_t(x_a^i)(1 - y_t(x_n^i))) \quad (3)$$

where N_{BN} is the batch size.

Subsequently, the ATM adjusts the teacher features by estimating the normalization statistics μ_{BN} and σ_{BN} and optimizing the transformation parameters γ_{BN} and β_{BN} , as depicted in Fig. 4. Specifically, normalization centers and standardizes the input x into $\bar{x} = (x - \mu_{BN}) / \sigma_{BN}$ using its mean μ_{BN} and standard deviation σ_{BN} . The transformation converts $\bar{x}$ into the output $x' = \gamma_{BN}\bar{x} + \beta_{BN}$, where the affine parameters for scale γ_{BN} and shift β_{BN} are optimized based on the loss.

F. Normality Remembering Enhancement

The NRE encourages the student model to focus on the intricate features that currently mimic the teacher the least within normal image regions. We designate these hard features as p_n^h , and the NRE reformulates the general KD loss L_{gkd} as defined in (1), into a new expression

$$L_{nre} = \frac{1}{N_h} \sum_{l=1}^L \sum_{p_n^h}^{\mathcal{P}_n^h} \left(1 - \cos(\phi_{p_n^h}^{t,l}, \phi_{p_n^h}^{s,l})\right) \quad (4)$$

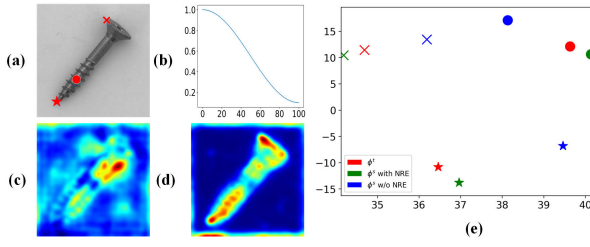


Fig. 5. Comparison with and without NRE. (a) Normal training image with sampling regions. (b) Decreasing curve of the mining factor h . (c) and (d) Anomaly score maps, respectively, without and with NRE at the tenth training epoch. (e) t-SNE-generated projection of feature vectors within the sampling regions [38].

where N_h is the total number of hard features. To identify p_n^h , we calculate the h -quantile of the cosine similarity values between the teacher and student features. Specifically, the h -quantile represents the h th smallest cosine similarity value among all the teacher–student feature pairs. This h -quantile threshold, labeled as d_h , allows for the computation of the loss L_{nre} as the average of all the distances $D(\phi_{p_n}^t, \phi_{p_n}^s)$ for which $\cos(\phi_{p_n}^t, \phi_{p_n}^s) < d_h$. Furthermore, we introduce a progressively decreasing factor $h \in [0, 1]$, defined as

$$h = 1 + (h_f - 1) \left(1 - \cos \left(\frac{\pi n_E}{N_E} \right) \right) / 2. \quad (5)$$

Here, n_E indicates the current training epoch index, N_E the total number of training epochs, and h_f (the final value of h) is empirically set to 0.001. The diminishing trajectory of h is depicted in Fig. 5(b). The gradual decline of h ensures a broader scope of feature distillation during the initial stages of training, which contributes to a more stable training progression.

Fig. 5(c) and (d) showcases a comparative analysis at the tenth training epoch, contrasting the anomaly maps without and with the application of NRE, respectively. Notably, the inclusion of NRE enables the student model to swiftly concentrate on intricate features, including the screw thread ($\bullet$), screw head at the tip ($\star$), and tail edge ($\times$), while effectively suppressing the unrelated background. Then the subsequent training epochs necessitate further refinement for the student model to harmonize challenging normal features with those of the teacher model. In addition, Fig. 5(d) depicts the t-SNE-based projection [38] of feature vectors after training within the sampled intricate regions, where red denotes $\phi_{p_n}^{t,1}$, green represents $\phi_{p_n}^{s,1}$ enhanced by NRE, and blue corresponds to $\phi_{p_n}^{s,1}$ without NRE. This visualization underscores the capacity of NRE to facilitate closer alignment of intricate features between the student and teacher models. Consequently, we posit that hard feature distillation by NRE can enhance the student model’s ability to recognize normals, thereby reducing the frequency of false-positive detections.

G. Direction-Guided Regularization

The DGR uses a feature direction regularization technique [39] to explicitly amplify the divergence of abnormal feature pairs between the student and teacher models.

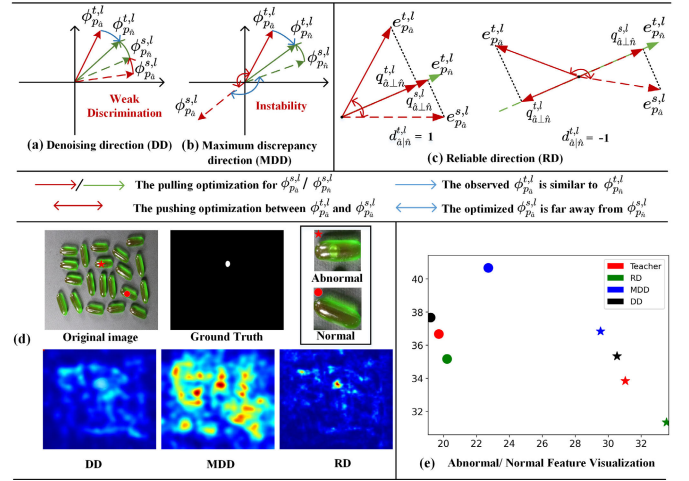


Fig. 6. Illustration of three types of DGR. (a) DD. (b) MDD. (c) RD. (d) Qualitative comparison with an abnormal sample. The anomaly maps are provided. (e) t-SNE projection of feature vectors within both an abnormal region ($\star$) and a normal region ($\bullet$).

Recently, two principal forms of DGR are identified: denoising direction (DD) and maximum discrepancy direction (MDD). This study examines the inherent limitations of both DD and MDD and introduces an innovative RD approach to counteract these weaknesses.

1) *Denoising Direction*: In the DD approach, the synthetic anomaly regions are treated as noise, prompting the student network to denoise within the feature space. While previous studies like DeSTSeg [20] have implemented denoising via an encoder–decoder network, we redefine DD from a regularization perspective. Specially, both the normal image x_n and its perturbed version $x_{\hat{a}}$ are fed into the student network. The DD regularization loss is then computed as follows:

$$L_{dd} = \frac{1}{N_{\hat{a}}} \sum_{l=1}^L \sum_{p_{\hat{a}}}^{\mathcal{P}_{\hat{a}}} \left(1 - \cos(\phi_{p_{\hat{a}}}^{s,l}, \phi_{p_{\hat{a}}}^{s,l}) \right) \quad (6)$$

where $p_{\hat{a}}$ represents the position of the simulated abnormal features, while $p_{\hat{n}}$ corresponds to normal features, both determined by the synthetic anomaly mask m .

L_{dd} aims to align $\phi_{p_{\hat{a}}}^{s,l}$ with $\phi_{p_{\hat{n}}}^{s,l}$. For distinct anomalies, $\phi_{p_{\hat{a}}}^{t,l}$ differs from $\phi_{p_{\hat{n}}}^{t,l}$. Since $\phi_{p_{\hat{a}}}^{s,l}$ is optimized to mimic $\phi_{p_{\hat{n}}}^{t,l}$, we expect $\phi_{p_{\hat{a}}}^{s,l}$ to also exhibit differences from $\phi_{p_{\hat{a}}}^{t,l}$. However, in cases of subtle anomalies, $\phi_{p_{\hat{a}}}^{t,l}$ may resemble $\phi_{p_{\hat{n}}}^{t,l}$, leading to weak discrimination due to the approximation sequence $\phi_{p_{\hat{a}}}^{s,l} \rightarrow \phi_{p_{\hat{a}}}^{s,l} \rightarrow \phi_{p_{\hat{n}}}^{t,l} \rightarrow \phi_{p_{\hat{a}}}^{t,l}$. Fig. 6(a) demonstrates this issue of weak discrimination, and Fig. 6(d) DD presents a case of Capsules where a subtle anomaly is not detected.

2) *Maximum Discrepancy Direction*: In the MDD approach, the objective is to directly optimize $\phi_{p_{\hat{a}}}^{s,l}$ to attain the maximum distance from $\phi_{p_{\hat{n}}}^{s,l}$, echoing the strategy used in CDO [15] and Pull&Push [16]. Specially, $x_{\hat{a}}$ undergoes simultaneous processing by both the teacher and student networks, resulting in the MDD loss function defined as

follows:

$$L_{\text{mdd}} = \frac{1}{N_{\hat{a}}} \sum_{l=1}^{\mathcal{L}} \sum_{p_{\hat{a}}}^{P_{\hat{a}}} \left(\cos(\phi_{p_{\hat{a}}}^{t,l}, \phi_{p_{\hat{a}}}^{s,l}) \right). \quad (7)$$

Compared with DD, MDD offers superior discrimination capabilities. However, in cases of subtle anomalies, MDD may induce significant deviations between $\phi_{p_{\hat{a}}}^{s,l}$ and $\phi_{p_{\hat{n}}}^{s,l}$. This, in turn, may cause training instability and heighten sensitivity to normal patterns. This instability is illustrated in Fig. 6(b), while Fig. 6(d) MDD showcases an instance where numerous delicate normal structures are erroneously detected, primarily due to this heightened sensitivity.

3) *Reliable Direction*: To address the drawbacks of both DD and MDD, we devise the RD regularization technique, which seeks to uphold MDD's high discriminative capability while incorporating a directional constraint guided by the similarity between $\phi_{p_{\hat{a}}}^{t,l}$ and $\phi_{p_{\hat{n}}}^{t,l}$. Initially, we normalize the feature vectors of $\phi_{p_{\hat{a}}}^{t,l}$, $\phi_{p_{\hat{a}}}^{s,l}$, and $\phi_{p_{\hat{n}}}^{t,l}$ to obtain $e_{p_{\hat{a}}}^{t,l}$, $e_{p_{\hat{a}}}^{s,l}$, and $e_{p_{\hat{n}}}^{t,l}$, respectively. Subsequently, we calculate the projections of $e_{p_{\hat{a}}}^{t,l}$ and $e_{p_{\hat{a}}}^{s,l}$ onto $e_{p_{\hat{n}}}^{t,l}$ as $q_{\hat{a}\perp\hat{n}}^{t,l} = e_{p_{\hat{a}}}^{t,l} \cdot e_{p_{\hat{n}}}^{t,l}$ and $q_{\hat{a}\perp\hat{n}}^{s,l} = e_{p_{\hat{a}}}^{s,l} \cdot e_{p_{\hat{n}}}^{t,l}$. Then, we determine the direction consistency between $e_{p_{\hat{a}}}^{t,l}$ and $e_{p_{\hat{n}}}^{t,l}$ using $d_{\hat{a}\perp\hat{n}}^{t,l} = \text{sign}(\cos(e_{p_{\hat{a}}}^{t,l}, e_{p_{\hat{n}}}^{t,l}))$. Finally, we formulate the RD loss as follows:

$$L_{\text{rd}} = \frac{1}{N_{\hat{a}}} \sum_{l=1}^{\mathcal{L}} \sum_{p_{\hat{a}}}^{P_{\hat{a}}} \left(\overbrace{\cos(e_{p_{\hat{a}}}^{t,l}, e_{p_{\hat{a}}}^{s,l})}^{\text{MDD}} + \overbrace{\|q_{\hat{a}\perp\hat{n}}^{t,l} - d_{\hat{a}\perp\hat{n}}^{t,l} \cdot q_{\hat{a}\perp\hat{n}}^{s,l}\|_2}^{\text{PC}} \right) \quad (8)$$

where PC stands for projection constraint. The mechanism of RD is illustrated in Fig. 6(c). For subtle anomalies where $\phi_{p_{\hat{a}}}^{t,l}$ and $\phi_{p_{\hat{n}}}^{t,l}$ align in the same direction, RD encourages divergence between $e_{p_{\hat{a}}}^{s,l}$ and $e_{p_{\hat{a}}}^{t,l}$ in the mirror direction, thereby mitigating the instability observed in MDD. Conversely, for larger anomalies where these vectors are oriented oppositely, RD optimizes divergence along the direction of maximum discrepancy, thereby preserving MDD's discriminative capacity.

Fig. 6(d) depicts the anomaly maps generated by DD, MDD, and RD, respectively. Notably, RD-based regularization demonstrates an ability to discern subtle anomalies more effectively than DD, while mitigating sensitivity toward delicate normal structures, when compared with MDD. In addition, Fig. 6(e) provides a comparative visualization of features for both the abnormal and normal regions, using t-SNE [38]. In the abnormal region, the student features for RD (★) and MDD (★) are more distant from the corresponding teacher feature (★) than the student feature for DD (★). Conversely, for the normal region, the student features for RD (●) and DD (●) are closer to the teacher feature (●) than the student feature for MDD (●). This comparative analysis underscores the proficiency of RD-based DGR in addressing the inherent limitations of both DD and MDD.

Finally, the total loss of REDAD is then expressed as

$$L_{\text{redad}} = L_{\text{ce}} + \alpha L_{\text{nre}} + \beta L_{\text{rd}} \quad (9)$$

where α and β serve as regularizers.

Algorithm 1 REDAD Pseudocode, PyTorch-Like

Input: T-S Models (ϕ^t, θ^t) and (ϕ^s, θ^s)
Output: (ϕ^t, θ^t), (ϕ^s, θ^s) and Anomaly Map S_{am}

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1 Train: Load ( $\phi^t, \theta^t$ ) and Initialize ( $\theta^s$ )
2 for  $x_n \leftarrow \text{dataloader}(\mathcal{X}_{\text{train}})$  do
3    $x_{\hat{a}}, m = \text{Perlin}(x_n) \Leftrightarrow (2)$ 
4    $L_{\text{ce}} = \text{CE}(y_t(x_{\hat{a}}), y_t(x_n)) \Leftrightarrow (3)$ 
5    $L_{\text{nre}} = \text{NRE}(\phi_{p_n}^t, \phi_{p_n}^s) \Leftrightarrow (4)$ 
6    $\phi_{p_{\hat{a}}}^t, \phi_{p_{\hat{a}}}^s, \phi_{p_{\hat{n}}}^t = m \odot \phi^t(x_{\hat{a}}), m \odot \phi^s(x_{\hat{a}}), m \odot \phi^t(x_{\hat{n}})$ 
7    $e_{p_{\hat{a}}}^t, e_{p_{\hat{a}}}^s, e_{p_{\hat{n}}}^t = \text{Normalize}(\phi_{p_{\hat{a}}}^t, \phi_{p_{\hat{a}}}^s, \phi_{p_{\hat{n}}}^t)$ 
8    $L_{\text{rd}} = \text{DGR}(e_{p_{\hat{a}}}^t, e_{p_{\hat{a}}}^s, e_{p_{\hat{n}}}^t) \Leftrightarrow (8)$ 
9    $L_{\text{redad}} = L_{\text{ce}} + \alpha L_{\text{nre}} + \beta L_{\text{rd}} \Leftrightarrow (9)$ 
10  Backward ( $L_{\text{redad}}$ ) and Update ( $\theta^s$ ), ATM ( $\theta^t$ )
11 end
12 Test: Load ( $\phi^t, \theta^t$ ) and ( $\phi^s, \theta^s$ )
13 for  $x_i \leftarrow \text{dataloader}(\mathcal{X}_{\text{test}})$  do
14    $S_{\text{am}} = \text{CalcAM}(\phi^t(x_i), \phi^s(x_i)) \Leftrightarrow (10)$ 
15 end

```

H. Anomaly Score Calculation

In the inference stage, we feed each $x_i \in \mathcal{X}_{\text{test}}$ into both the teacher and student networks. The multihierarchical features $\phi^{t,l}$ and $\phi^{s,l}$ are extracted and resized to the same spatial resolution as the first-layer feature map (H_1, W_1). Then the fused anomaly score can be calculated as

$$s_p = \sum_{l=1}^{\mathcal{L}} (1 - \cos(\phi_p^{t,l}, \phi_p^{s,l})). \quad (10)$$

The anomaly score map $\{s_p | p \in (H_1, W_1)\}$ is interpolated to have the resolution of the input image size (H_0, W_0) and Gaussian filtered for smooth boundaries.

For the anomaly localization task, we can obtain the anomaly mask with a threshold τ by

$$S_{\text{AL}}(x_i) := \{p | s_p > \tau, p \in H_0 \times W_0\}. \quad (11)$$

For the image-level AD task, the maximum score of the anomaly map is taken as the AD score

$$S_{\text{AD}}(x_i) := \max_{p \in H_0 \times W_0} s_p. \quad (12)$$

The pseudocode for the train and test procedures of REDAD is shown in Algorithm 1, where θ is the model parameter.

IV. EXPERIMENTS

A. Implementation Details

1) *Dataset Descriptions*: The MVTec AD [1] comprises 15 datasets for AD, encompassing ten object classes and five texture types, and serves as a widely recognized standard for benchmarking AD algorithms. The anomalies within these datasets primarily consist of various structural changes. The more recent VisA dataset [40] extends this concept by incorporating a diverse array of objects characterized by complex structures, multiple instances, and variations in scale. VisA represents the largest open-source industrial AD dataset to date, comprising 10821 high-resolution color images of

TABLE I

QUANTITATIVE RESULTS IN TERMS OF I-AUROC, P-AUROC, AND PRO ON THE MVTEC [1] DATASETS. AVG. T., AVG. O., AND AVG. ALL ARE THE AVERAGE RESULTS OVER THE TEXTURE, OBJECTIVE, AND ALL CATEGORIES, RESPECTIVELY. THE TOP PERFORMANCE IS HIGHLIGHTED IN BOLD

Method	CFlow			ReConPatch			DDAD			STPM			MemKD			CDO			Pull&Push			DeSTSeg			EfficientAD			REDAD		
Carp.	100.0	99.3	97.7	99.6	98.8	-	99.3	98.7	93.9	98.1	99.1	95.6	-	-	-	98.2	99.1	96.8	95.9	99.5	98.3	99.0	96.1	96.0	-	-	-	100.0	99.4	98.2
Grid	97.6	99.0	96.1	100.0	99.0	-	100.0	99.4	97.3	97.4	99.1	96.6	-	-	-	99.8	98.4	96.0	99.9	99.4	97.7	100.0	99.1	97.1	-	-	-	100.0	99.4	97.8
Lea.	97.7	99.7	99.4	100.0	96.0	-	100.0	99.4	97.7	100.0	99.2	98.9	-	-	-	100.0	99.2	98.3	63.6	99.7	98.7	100.0	99.7	99.1	-	-	-	100.0	99.6	98.9
Tile	98.7	98.0	94.3	98.8	98.9	-	100.0	98.2	93.1	95.4	96.6	89.9	-	-	-	99.5	97.2	90.5	99.7	96.8	90.3	100.0	98.0	93.3	-	-	-	99.8	98.5	94.0
Wood	99.6	96.7	95.8	99.7	98.9	-	100.0	95.0	82.9	98.3	95.2	95.2	-	-	-	99.7	95.9	92.9	99.6	95.2	93.2	97.8	97.7	95.8	-	-	-	100.0	98.2	94.6
Avg. T.	98.7	98.5	96.7	99.8	98.3	-	99.9	98.1	93.0	97.8	97.8	95.2	-	-	-	99.4	98.0	94.8	91.7	98.1	95.6	99.4	98.1	96.3	-	-	-	100.0	99.0	96.7
Bottle	100.0	99.0	96.8	100.0	98.2	-	100.0	98.7	91.8	99.8	98.8	94.4	-	-	-	100.0	99.3	97.2	99.9	98.7	95.6	100.0	99.2	96.3	-	-	-	100.0	99.1	96.5
Cable	100.0	97.6	93.5	99.8	99.3	-	99.4	98.1	88.9	96.1	94.8	88.7	-	-	-	96.8	97.6	94.2	98.4	95.7	87.5	98.0	97.3	84.8	-	-	-	98.5	97.2	93.7
Caps.	99.3	99.0	93.4	98.8	97.6	-	99.4	95.7	93.4	90.6	98.2	88.3	-	-	-	95.6	98.6	93.0	99.8	97.8	89.9	96.5	99.1	94.5	-	-	-	98.2	98.9	94.8
Hazel.	96.8	98.9	96.7	100.0	98.9	-	100.0	98.4	86.7	100.0	98.9	96.3	-	-	-	100.0	99.2	97.4	99.5	98.6	96.1	99.9	99.6	97.9	-	-	-	100.0	99.5	97.5
M.nut	91.9	98.6	91.7	100.0	95.8	-	100.0	99.0	91.1	99.2	97.2	94.3	-	-	-	99.2	98.5	95.7	86.9	97.8	93.2	99.5	98.6	94.4	-	-	-	100.0	98.5	95.9
Pill	99.9	99.0	95.4	97.5	95.3	-	100.0	99.1	95.5	91.8	94.7	96.2	-	-	-	97.9	98.9	96.6	99.7	98.6	94.9	97.6	98.7	90.3	-	-	-	98.2	98.3	97.9
Screw	99.7	98.9	95.3	98.5	98.8	-	99.0	99.3	96.3	85.4	98.6	91.1	-	-	-	90.8	99.0	94.3	85.3	96.9	85.6	91.2	98.5	93.8	-	-	-	97.6	99.5	95.6
T.brush	95.2	99.0	95.1	100.0	98.9	-	100.0	98.7	92.6	96.1	98.9	92.3	-	-	-	91.2	98.9	90.5	94.0	99.1	91.8	99.2	99.3	93.6	-	-	-	98.6	98.5	93.1
Trans.	99.1	98.0	81.4	100.0	99.7	-	100.0	95.3	90.1	94.0	81.9	76.7	-	-	-	96.5	95.3	92.6	100.0	99.2	97.6	98.1	89.1	90.8	-	-	-	98.8	97.2	93.4
Zipper	98.5	99.1	96.6	99.8	98.6	-	100.0	98.2	93.2	95.3	98.0	95.3	-	-	-	98.7	98.2	94.3	99.6	97.9	93.0	100.0	99.1	97.8	-	-	-	99.4	99.1	96.5
Avg. O.	98.0	98.7	93.6	99.4	98.1	-	99.9	98.0	92.0	94.8	96.0	91.3	-	-	-	96.7	98.4	94.6	96.3	98.0	92.5	98.0	97.8	93.4	-	-	-	98.9	98.6	95.5
Avg. All	98.3	98.6	94.6	99.6	98.2	-	99.8	98.1	92.3	95.8	96.6	92.7	99.6	98.2	94.5	97.6	98.2	94.7	94.8	98.1	93.6	98.5	97.9	94.4	99.1	-	93.5	99.3	98.7	95.9

12 different objects. The 3-D dataset MVTEC3D AD [41] creates a task to localize mainly geometric anomalies in 3-D objects. MVTEC3D includes 4147 scans captured by a 3-D sensor from ten object categories, each providing both RGB and point cloud data. When localizing geometry anomalies in MVTEC3D, we do not use 3-D scans, as the proposed REDAD only requires RGB images as inputs.

2) *Experimental Settings*: We used the WResNet50 [42] as the backbone in the S-T models, aiming for a balanced tradeoff between effectiveness and efficiency. We configured $\mathcal{L} = 3$, enabling the extraction of feature maps from the outputs of the first three hierarchies. All the images in both the training and testing datasets were resized to 256×256 . For data augmentation, we exclusively applied normalization with the ImageNet mean and variance to the input images. For optimization, we used the AdamW optimizer with a learning rate of 0.01 and a batch size of 32. In addition, we adopted an exponential moving average (EMA) [43] with a decay rate of 0.9 to update the student model parameters. The training process encompassed 100 epochs. Regarding the loss function (9), we assigned weights of 1 and 0.1 to α and β parameters, respectively. The threshold τ in (11) was determined as the globally optimal value based on the $F1$ -score, facilitating the accurate segmentation of abnormal regions in qualitative evaluations. Finally, we used a Gaussian filter with a standard deviation of $\sigma = 4$ to smooth the anomaly score map, as described in Section III-H. Model training and testing were conducted on a computer equipped with an Intel¹ Xeon¹ CPU and an NVIDIA Tesla V100 GPU.

3) *Evaluation Metrics*: We adopted the area under the receiver operating characteristic curve (AUROC) based on produced anomaly scores to calculate AD at image-level performance (I-AUROC). Localization performance was evaluated using the AUROC at pixel level (P-AUROC) and PRO [8]. For I-AUROC, P-AUROC, and PRO, a higher value indicates better detection performance.

B. Comparisons to State-of-the-Art Methods

We conducted a comparative analysis of REDAD's performance against several prominent competitors, including CFlow [28], ReConPatch [27], DDAD [24], STPM [10], MemKD [13], CDO [15], Pull&Push [16], DeSTSeg [20], and EfficientAD [14]. Notably, CFlow exemplifies the normalizing flow approach, while ReConPatch and DDAD represent the cutting-edge memory bank-based and reconstruction-based methodologies, respectively. STPM serves as the benchmark for general KDST-based methods. CDO, Pull&Push, and DeSTSeg are recognized as leading KDST-based approaches, relevant to our DGR method, as elaborated in Section III-G. EfficientAD combines KDST and reconstruction methods, while MemKD integrates KDST and memory bank methods, both achieving state-of-the-art performance. It is worth noting that ReConPatch, MemKD, Pull&Push, and EfficientAD are not publicly available, whereas the remaining comparison methods are open-sourced.

As detailed in Tables I and II, REDAD exhibits superior performance across both the MVTEC and VisA datasets. Specifically, REDAD achieved notable improvements over the baseline STPM, with a 3.5% enhancement in I-AUROC, a 2.3% increase in P-AUROC, and a 3.2% boost in PRO for the MVTEC dataset, and a 5.0% gain in I-AUROC, a 1.6% rise in P-AUROC, and a 5.4% improvement in PRO for the VisA dataset. When compared with its closest competitors, including CDO, Pull&Push, and DeSTSeg, REDAD demonstrates superior anomaly image classification and anomaly pixel localization capabilities. Furthermore, in comparison to state-of-the-art methods such as ReConPatch, DDAD, MemAD, and EfficientAD, REDAD achieves comparable performance in I-AUROC and the highest scores in P-AUROC. For the 3-D AD dataset MVTEC3D, we only compared REDAD with STPM, MemKD, and CDO, as these were the only methods that had tested the MVTEC3D dataset in their work. It can be observed in Table III that REDAD outperforms STPM by 6.1% in I-AUROC, 1.5% in P-AUROC, and 3.8% in PRO. In addition, REDAD surpasses CDO in all three metrics.

¹Registered trademark.

TABLE II

QUANTITATIVE RESULTS IN TERMS OF I-AUROC, P-AUROC, AND PRO ON THE VISA [40] DATASETS. THE TOP PERFORMANCE IS HIGHLIGHTED

Method	CFlow			ReConPatch			DDAD			STPM			MemKD			CDO			Pull&Push			DeSTSeg			EfficientAD			REDAD			
Candle	95.9	98.7	93.0	-	-	-	99.9	98.7	96.6	83.3	97.4	92.8	-	-	-	93.6	99.2	95.4	-	-	-	96.2	98.3	95.1	-	-	-	95.7	99.0	95.5	
Caps.	82.1	97.7	88.0	-	-	-	100.0	99.5	95.0	86.6	96.1	90.7	-	-	-	91.8	99.1	91.3	-	-	-	85.2	95.5	89.5	-	-	-	96.4	99.6	93.5	
Cashew	93.6	99.2	92.2	-	-	-	94.5	97.4	80.3	91.2	92.5	79.4	-	-	-	93.4	98.7	94.9	-	-	-	93.1	97.6	85.1	-	-	-	93.1	98.1	93.8	
C.gum	99.6	99.1	98.5	-	-	-	98.1	96.5	85.2	95.2	98.5	74.4	-	-	-	98.6	99.1	87.4	-	-	-	96.0	97.8	86.1	-	-	-	99.9	99.5	91.4	
Fryum	93.1	97.4	78.4	-	-	-	99.0	96.9	94.2	86.6	95.1	85.0	-	-	-	94.9	95.1	84.8	-	-	-	92.2	95.4	87.2	-	-	-	98.7	96.8	92.6	
Maca.1	89.6	98.7	89.3	-	-	-	99.2	98.7	98.5	90.4	99.5	94.8	-	-	-	94.5	99.6	95.9	-	-	-	86.6	99.1	95.1	-	-	-	94.9	99.5	96.7	
Maca.2	85.3	97.9	85.8	-	-	-	99.2	98.2	99.3	83.9	99.2	94.1	-	-	-	89.6	99.5	95.4	-	-	-	80.9	99.2	97.1	-	-	-	95.4	98.6	95.0	
Pcb1	95.5	99.2	89.1	-	-	-	100.0	93.4	93.3	93.4	99.4	93.3	-	-	-	90.7	99.6	94.4	-	-	-	86.1	96.1	91.0	-	-	-	96.1	99.8	95.5	
Pcb2	89.5	95.6	85.1	-	-	-	99.7	97.4	93.3	93.7	98.4	89.2	-	-	-	90.7	98.3	87.1	-	-	-	93.4	98.5	89.1	-	-	-	95.9	99.0	93.4	
Pcb3	78.2	96.9	80.0	-	-	-	-	97.2	96.3	86.6	91.8	98.5	88.9	-	-	-	95.1	98.6	89.7	-	-	-	98.0	98.4	92.1	-	-	-	95.3	99.2	94.2
Pcb4	94.1	97.1	83.7	-	-	-	100.0	98.5	95.5	98.2	98.5	91.2	-	-	-	96.1	98.9	92.4	-	-	-	98.3	96.1	85.1	-	-	-	98.6	98.3	92.5	
P.fryum	93.6	98.8	91.7	-	-	-	100.0	99.5	94.7	90.7	98.2	90.4	-	-	-	98.2	99.2	94.1	-	-	-	96.9	99.7	95.2	-	-	-	96.4	98.2	94.8	
Avg.	90.8	98.0	87.9	-	-	-	98.9	97.6	92.7	90.4	97.6	88.7	97.6	98.4	94.9	93.9	98.7	91.9	-	-	-	91.9	97.6	90.6	98.1	-	94.0	96.4	98.8	94.1	

TABLE III

QUANTITATIVE RESULTS IN TERMS OF I-AUROC, P-AUROC, AND PRO ON THE MVTEC3D [41] DATASETS. THE TOP PERFORMANCE IS HIGHLIGHTED

Method	STPM			MemKD			CDO			REDAD		
Bagel	94.5	98.4	93.2	99.1	-	-	98.5	98.6	97.5	94.8	98.8	94.7
C.Gland	90.3	95.2	91.0	95.3	-	-	93.3	99.2	98.3	95.7	96.0	96.3
Carrot	82.1	96.9	97.5	95.7	-	-	91.9	98.5	98.1	93.0	97.2	98.9
Cookie	63.1	96.3	91.7	71.6	-	-	68.8	96.8	86.3	77.3	98.2	91.2
Dowel	87.8	97.3	89.8	99.3	-	-	96.2	99.7	97.6	92.3	98.5	97.4
Foam	85.2	95.8	69.8	87.5	-	-	75.0	91.3	70.5	86.6	98.5	82.1
Peach	89.4	96.3	93.3	95.0	-	-	91.9	99.5	98.6	96.9	97.9	97.2
Potato	73.4	94.5	95.5	67.9	-	-	76.9	99.2	96.1	80.1	98.4	93.7
Rope	94.6	99.1	94.5	99.8	-	-	97.1	98.6	97.1	99.9	99.5	94.7
Tire	90.6	97.2	88.4	88.9	-	-	90.1	98.6	97.4	95.8	99.5	96.8
Avg. All	85.1	96.7	90.5	90.0	-	-	88.0	98.0	93.8	91.2	98.2	94.3

This exceptional performance in anomaly localization can be attributed to two key aspects of our proposed method. First, the NRE module enables the model to reinforce intricate feature learning between the student and teacher models, thereby reducing false alarms. Second, our DGR technique effectively controls the distillation of synthetic anomalies between the teacher and student models, thereby endowing the model with high discriminative power for anomalies. In other words, when facing scenes with complex detailed structures and subtle defects, REDAD often outperforms other competitors.

To underscore the exceptional performance of REDAD, Fig. 7 displays qualitative outcomes from selected samples across the MVTEC, Visa, MVTEC3D, and our curated datasets of vacuum flasks and wheel hubs. These results highlight REDAD's consistent efficacy in identifying a broad spectrum of defects. This achievement is attributed to REDAD's ability to simultaneously address the prevalent challenges within the KDST framework, including data shift, normality forgetting, and anomaly overgeneralization, thereby ensuring its effectiveness across various industrial applications.

C. Issues' Mitigation

We assessed the effectiveness of the proposed ATM, NRE, and DGR modules in addressing issues present in general KDST-based methods. The MVTEC dataset is used, and the texture and object categories are analyzed separately. Based

on the basic setting in (I), we integrated each of the three modules separately, with the results depicted in Fig. 8.

1) *Data Shift*: As illustrated in Fig. 8(a), the implementation of ATM led to a notable reduction in KD loss. This is due to the domain adaptation for the feature maps of the teacher model by BN updating in ATM, and then they can be highly aligned with the feature maps of the student model which is learned using the MVTEC datasets. The texture category, specifically Grid, experienced a more significant decrease compared with the object category, Bottle. This disparity is attributed to texture category images being less similar to the images in the pretrained ImageNet dataset than those in the object category [see Fig. 7(a)].

2) *Normality Forgetting*: To verify the effectiveness of NRE in remembering the normality, we analyzed the cumulative histogram distribution (CHD) of normal feature similarity between the teacher and student networks using the testing datasets. Fig. 8(b) shows that the inclusion of NRE resulted in a greater proportion of high similarity feature pairs than it without NRE, indicating enhanced consistency of normality.

3) *Overgeneralization*: To verify the effectiveness of DGR in overcoming anomaly overgeneralization, we analyzed the CHD of anomaly feature similarity between the teacher and student networks using testing datasets. For the abnormal regions, Fig. 8(c) clearly demonstrates that the application of DGR shifted the model's focus toward lower similarity, indicating a heightened level of discrepancy.

D. Ablation Studies

We conducted ablation studies to assess the individual contributions of REDAD's components, and the outcomes using the MVTEC, Visa, and MVTEC3D datasets are presented in Table IV. By contrasting Case 1 with Cases 2, 3, and 4, it becomes evident that ATM, NRE, and RD individually exhibit enhanced performance over the baseline KDST. This observation underscores the efficacy of each component when evaluated in isolation. Furthermore, in comparing Case 3 with Case 4, our analysis reveals that NRE surpasses RD in image-level performance, whereas RD excels over NRE at the pixel level. We hypothesize that NRE's ability to efficiently address the challenges associated with normalcy retention, as discussed in Section III-F, significantly reduces false-positive detections, thereby enhancing image-level AD. Concurrently, the explicit

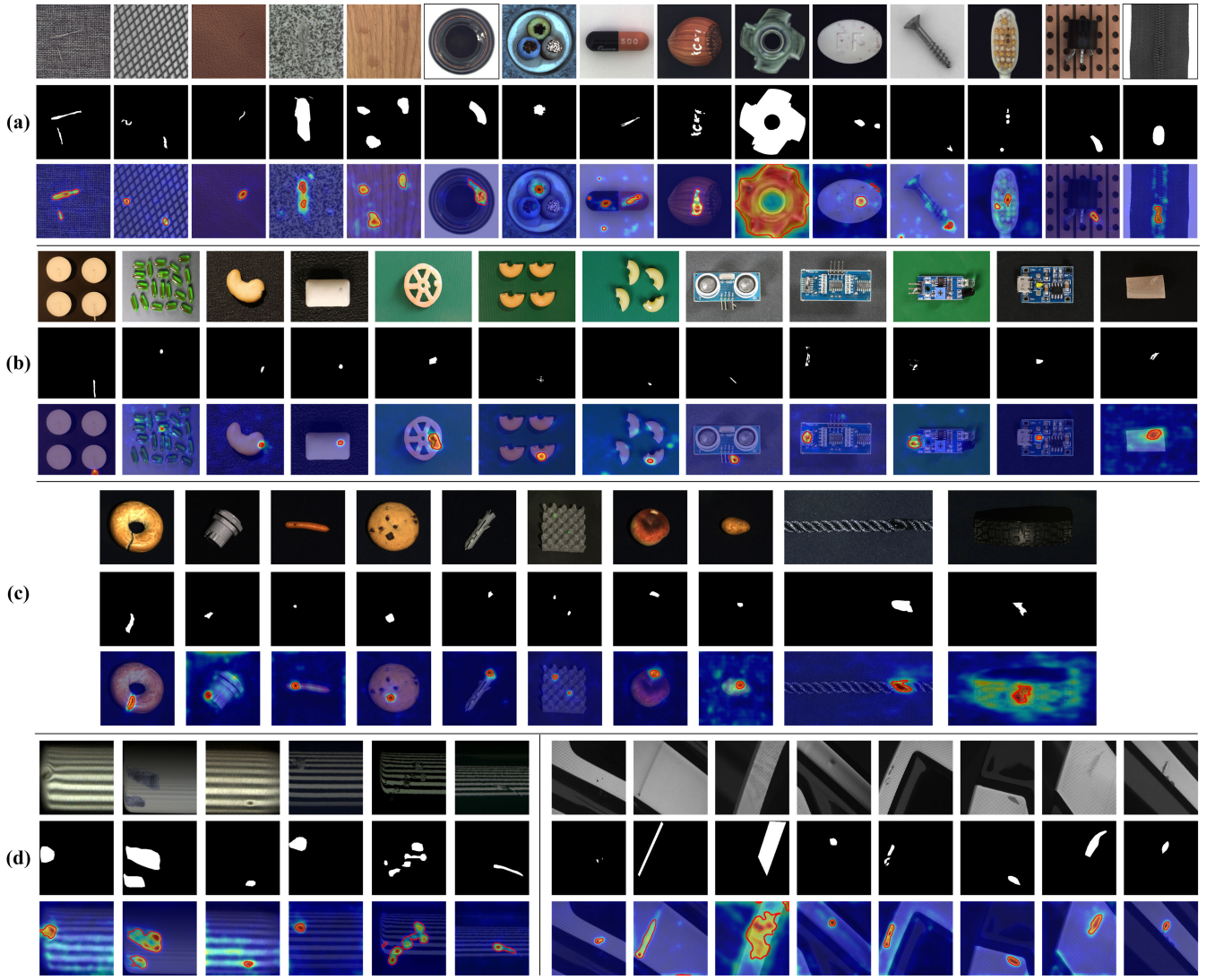


Fig. 7. Qualitative results of the proposed REDAD on (a) MVTec, (b) VisA, (c) MVTec3D, and (d) collected datasets of vacuum flasks and wheel hubs. Original images, ground-truth masks, and anomaly maps are listed from top to bottom.

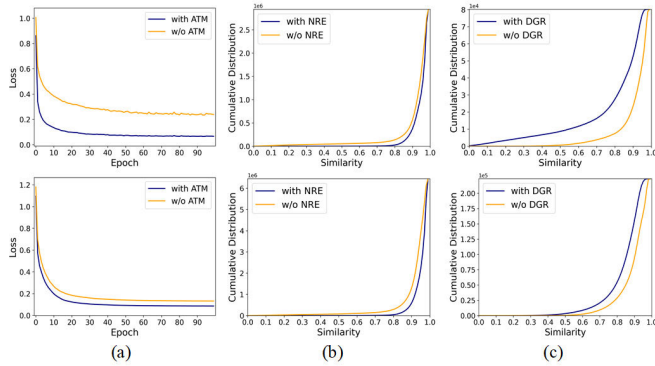


Fig. 8. Issues' mitigation verified using the MVTec dataset. The categories of texture (top) and objects (bottom) are analyzed separately. (a) Loss plots to verify the influence of ATM using grid (top) and bottle (bottom) categories. (b) Similarity statistics of normal regions to verify the influence of NRE. (c) Similarity statistics of abnormal regions to verify the influence of DGR.

utilization of synthetic anomalies through DGR effectively improves anomaly pixel localization. Ultimately, in Case 7,

TABLE IV
ABLATION STUDIES ON REDAD

Exp.	Component					MVTec			VisA			MVTec3D		
	ATM	NRE	DD	MDD	RD									
①						96.1	96.9	93.1	91.5	97.9	90.4	87.3	97.1	91.5
②	✓					97.3	97.1	93.3	92.1	97.9	91.2	89.5	97.4	92.6
③		✓				98.6	97.9	93.8	94.5	98.2	93.8	90.9	97.6	93.5
④					✓	97.9	98.3	94.1	94.0	98.4	94.0	90.6	98.0	94.1
⑤	✓	✓				98.9	98.1	94.0	95.6	98.0	93.8	91.1	97.8	93.6
⑥	✓	✓			✓	98.2	98.5	94.5	94.5	98.6	94.1	90.8	97.9	94.2
⑦	✓	✓			✓	99.3	98.7	95.9	96.4	98.8	94.1	91.2	98.2	94.3

our integrated REDAD approach attains superior performance, emphasizing the complementary nature and synergistic effects of the three modules. In addition, we evaluated three types of DGR methods—DD, MDD, and RD—in Cases 5, 6, and 7, respectively. The outcomes affirm that the proposed RD method outperforms others, reinforcing its reliability and efficacy.

E. Comparison of Different Backbones and Resolutions

The choice of backbone architecture and image resolution significantly affects the performance of localization tasks [28]. This research evaluates various backbones, such as ResNet18, ResNet34, ResNet50, ResNet101, and WResNet50, at two resolutions, 256×256 and 512×512 , using the MVTec dataset. The results are shown in Table V. Our findings indicate that WResNet50 at 256×256 offers the optimal balance between accuracy and efficiency. Larger models with increased parameters, such as ResNet101, even at 512×512 , yield minimal improvements or, in some cases, show no significant enhancement in performance. Consequently, for this study, we have selected WResNet50 operating at 256×256 .

F. Complexity Evaluation

This study assesses the computational and memory complexity of the proposed REDAD, focusing on inference speed and model size. STPM [10], CDO [15], Pull&Push [16], and REDAD share identical network architectures during inference; thus, their inference speeds and model sizes can be considered equivalent. In contrast, DeSTSeg [20] and MemKD [13] augment the fundamental student-teacher structure with additional components, theoretically resulting in slower inference times compared with methods based solely on the student-teacher framework. The inference speed of the diffusion-model-based DDAD [24] largely depends on the noise time step configuration, whereas ReConPatch [27], which uses a memory-bank approach, depends on the memory-bank's dimensions. Given these dependencies, neither DDAD nor ReConPatch is suitable for a fair comparative analysis. Consequently, we selected CFlow [28] and EfficientAD [14] as benchmarking counterparts.

The comparison of inference speeds and model sizes is shown in Table VI. The inference speed was measured on a computer with an Intel Xeon CPU and an NVIDIA Tesla V100 GPU. All the models were assigned to the GPU for inference acceleration, and the data input time was factored in during the evaluation of each method's inference speed. The model size is defined as the size of all the floating point parameters in the corresponding model. CFlow consists of a feature-extraction network and a decoder. Since the decoder is typically more complex, CFlow has a larger model size and a slower inference speed than REDAD. EfficientAD integrates a student-teacher architecture with a denoising decoder. Generally, this combination would result in a slower inference speed and a larger model size compared with REDAD. However, EfficientAD trains a lightweight patch description network (PDN) to boost its inference speed. As a result, EfficientAD demonstrates faster inference and a relatively efficient model size. Therefore, to further enhance REDAD's inference speed and reduce its memory footprint, adopting a smaller and more efficient backbone is a promising approach.

G. Parameters' Analysis

This study analyzes three critical hyperparameters: the anomaly localization threshold τ in (11), the hierarch param-

TABLE V
PERFORMANCE UNDER DIFFERENT BACKBONES AND RESOLUTIONS USING THE MVTEC DATASET

Backbone	256×256			512×512		
ResNet18	97.6	97.5	93.7	98.0	98.2	94.0
ResNet34	98.4	98.0	94.3	98.6	98.1	94.5
ResNet50	99.0	98.3	94.5	99.1	98.6	94.7
ResNet101	98.9	98.5	95.1	98.7	98.6	95.2
WResNet50	99.3	98.7	95.9	99.2	98.8	95.9

TABLE VI
COMPLEXITY COMPARISON IN TERMS OF INFERENCE SPEED (FPS) AND MODEL SIZE (MB)

Method	Backbone	Resolution	Inference Speed	Model Size
REDAD	ResNet18	256×256	26.21	89
	WResNet50	256×256	24.53	518
CFlow	ResNet18	256×256	14.72	96
	WResNet50	256×256	12.37	902
EfficientAD	PDN-S	256×256	62.64	100
	PDN-M	256×256	36.43	161

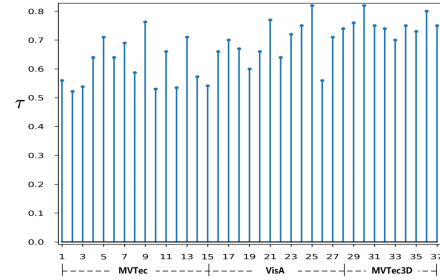


Fig. 9. Thresholds established using the globally optimal $F1$ -score for each category across the test datasets of MVTec, VisA, and MVTec3D.

eter $\mathcal{L}$, and the h -quantile in (5). The metrics I-AUROC, P-AUROC, and PRO are threshold-agnostic, thereby avoiding bias in model performance evaluation. However, in practical applications, setting a threshold is necessary to obtain final AD results. Fig. 9 illustrates the thresholds established using the globally optimal $F1$ -score for each category across the MVTec, VisA, and MVTec3D test datasets. It can be observed that the optimal thresholds relative to the $F1$ -score fall within the interval $[0.5, 0.8]$. Notably, the optimal thresholds for MVTec3D lie within a narrower and higher range of $[0.7, 0.8]$, compared with those of MVTec and VisA. This phenomenon can be attributed to the reduced background interference and simpler targets with more pronounced defects in the MVTec3D dataset [see Fig. 7(c)], leading to higher confidence in the anomaly scores produced by the model. Consequently, the selection of threshold τ in practical applications should consider specific scenarios. For instance, a high threshold, such as 0.6 or 0.7, can ensure high precision in settings with clean backgrounds and clear defects. Conversely, a lower threshold, like 0.4 or 0.5, may be appropriate for complex backgrounds or subtle defects to maintain high recall.

The hierarch parameter $\mathcal{L}$ also significantly influences model performance. Lower level feature maps capture detailed information from images, while higher level feature maps

TABLE VII
PERFORMANCE UNDER DIFFERENT HIERARCHS

Hierarchy	$\mathcal{L} = 1$			$\mathcal{L} = 2$			$\mathcal{L} = 3$			$\mathcal{L} = 4$			$\mathcal{L} = 5$		
MVTec	98.9	96.9	92.4	99.2	97.8	94.3	99.3	98.7	95.9	99.0	98.8	95.9	98.5	98.7	95.7
VisA	96.1	95.8	91.1	96.5	96.9	92.1	96.4	98.8	94.1	96.1	98.7	94.3	95.9	98.5	94.0
MVTec3D	90.7	96.8	90.7	91.1	97.5	92.5	91.2	98.2	94.3	91.0	98.3	94.3	90.8	98.2	94.2

TABLE VIII
PERFORMANCE UNDER DIFFERENT h -QUANTILE

Hierarchy	$h = 0.0$			$h = 0.1$			$h = 0.5$			$h = 1.0$			$h_{decline}$		
MVTec	72.3	69.5	61.3	97.5	96.9	92.7	98.4	98.5	95.0	98.0	98.2	94.4	99.3	98.7	95.9
VisA	69.5	71.2	66.7	93.6	96.5	92.1	95.9	98.7	94.0	94.2	98.1	93.7	96.4	98.8	94.1
MVTec3D	65.6	62.9	59.4	88.9	90.1	89.2	91.0	98.1	93.8	90.5	97.9	93.5	91.2	98.2	94.3

extract coarser semantic features. REDAD uses a pyramid structure to compare differences between teacher and student model feature maps at various levels for AD. Table VII presents the detection effectiveness under different hierarchs. Using only the top-level feature map ($\mathcal{L} = 1$) leads to a notable decline in performance on the MVTEC, VisA, and MVTEC3D datasets, as many fine-grained anomalies go undetected. Conversely, incorporating all the stage feature maps ($\mathcal{L} = 5$) does not improve performance and even causes a slight decrease due to increased noise from finer grained features. Therefore, we selected $\mathcal{L} = 3$ as the hierarchy parameter to balance detail and semantic information effectively.

The h -quantile of the cosine similarity values between the teacher and student features is used to identify the hard normal feature in the NRE module of the proposed REDAD. In REDAD, a gradual decline of h is implemented [see (5)]. To demonstrate the benefits of this gradual decline, we compared it with fixed values of h . The results are presented in Table VIII. Specifically, $h = 1.0$ indicates no hard example mining, while $h = 0$ implies no utilization of normal features for distillation. It is evident that when $h = 0$, the model performance is significantly degraded, highlighting the importance of distillation using normal features. Comparing $h = 1.0$ and $h = 0.1$ reveals that excessive hard example mining from the beginning of training leads to a decline in model detection performance. In contrast, comparing $h = 1.0$ and $h = 0.5$ shows that moderate hard example mining enhances performance. The gradual hard example mining approach adopted by REDAD achieves the best performance. This is because the gradual decline of h ensures a broader scope of feature distillation during the initial stages of training, thereby contributing to a more stable training progression.

H. Real-World Applications

For additional validation, this study has compiled two real-world industrial inspection datasets: vacuum flasks and aluminum alloy car wheel hubs, both meticulously annotated pixel-by-pixel. The used inspection equipment and exemplar images are showcased in Fig. 10. Comprehensive details regarding the datasets are provided in the central column of Table IX. The study has amassed a collection that includes 30 varieties of vacuum flasks with six types of defects and 20 types of wheel hubs with nine types of defects.

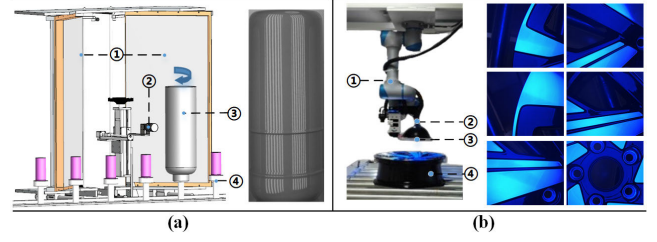


Fig. 10. Inspection of surface defects on vacuum flasks and car wheel hub. (a) Inspection device of vacuum flasks, ① stripe light sources, ② camera, ③ vacuum flask, and ④ rotating rod. (b) Inspection device of wheel hub, ① robotic arm, ② camera, ③ blue planar light source and reflector, and ④ wheel hub.

TABLE IX
DATASET DESCRIPTION AND PERFORMANCE ON VACUUM FLASKS AND WHEEL HUBS

Datasets	Types	Defect	#Normal	#Abnormal	Solution	Performance
Vacuum Flasks	30	6	10624	3547	2448×2048	97.8 98.2 95.7
Wheel Hubs	20	9	10012	2582	2448×2048	98.5 99.1 96.4

The image solution of both the datasets is 2448×2048 . To detect tiny defects, high-definition images were divided into 4×4 patches, which were then fed into the models during both the training and testing phases. We used 6000 normal images to train the models for both the vacuum flasks and wheel hubs. The rest of images were used for testing. The performance results are detailed in the rightmost column of Table IX, and the visual detection results are displayed in Fig. 7(c). The quantitative and qualitative evaluations demonstrate that the proposed REDAD method achieved impressive performance in AD.

In real industrial production lines, high-speed processing is essential due to the fast pace. Patch preprocessing, however, can lead to increased processing time. On the contrary, without patch preprocessing, it becomes challenging to detect minute defects. Therefore, the decision on how to patch must consider factors such as production speed, defect size, and hardware resources. In addition, compared with the existing AD datasets, our collected datasets exhibit significant interclass differences between patches within the image dataset. Taking wheel hubs as an example, different models of wheel hubs have distinct structures. As the number of product models increases, a substantial amount of normal images are required to train the model, which imposes a heavier training burden. Furthermore, the introduction of new wheel hub models necessitates retraining of the model, further complicating its deployment and application.

V. CONCLUSION

In this article, we introduce REDAD, a reliable image AD technique designed to address issues such as data shift, normality forgetting, and anomaly overgeneralization present in general KDST-based methods. REDAD consists of three components: ATM, NRE, and DGR, each targeting the respective issues. The effectiveness of the synergistic effect of these three modules has been validated through ablation experiments. Our method has undergone thorough testing on three benchmark datasets (MVTec, VisA, and MVTEC3D) for AD,

with outcomes showing considerable enhancements compared to other leading methods. Specifically, compared with the baseline model STPM [10], REDAD exceeds 3.5% (5.0%, 6.1%) in I-AUROC, 2.3% (1.6%, 1.5%) in P-AUROC, and 3.2% (5.4%, 3.8%) in PRO for the three datasets, respectively. In addition, we analyze the impact of model structure, input image resolution, number of hierarchical feature maps, and anomaly localization threshold on model performance, as well as conduct complexity evaluation, providing a more comprehensive evaluation. Furthermore, we validate REDAD on two collected real-world industrial product inspection datasets, confirming its effectiveness in practical applications.

Looking ahead, given the prevalence of data streams with frequent product changes in industrial manufacturing, we plan to incorporate REDAD into a continuous image AD learning framework to accommodate such dynamic industrial environments. In addition, to meet the demand for lower inference latency, we will further adopt a more lightweight backbone to improve inference efficiency while maintaining detection accuracy.

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